AI for Medical Imaging

Master Artificial Intelligence

Hoel Kervadec

Fall 2026

Universiteit van Amsterdam
Instituut voor Informatica

Amsterdam UMC

2026/09/08 – Registration

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Announcements

Announcements

Exam: Minimum passing grade of 5.5

Quizz from 2024 available on Canvas already

Rubric for project (non-final) available by end of the week

Tomorrow: nationwide public transport strike: meet your TAs online

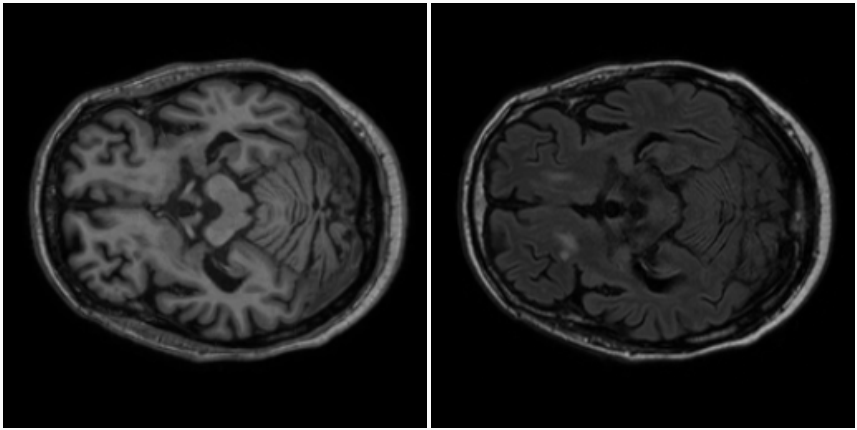
2026/09/08 – Registration

Registration?

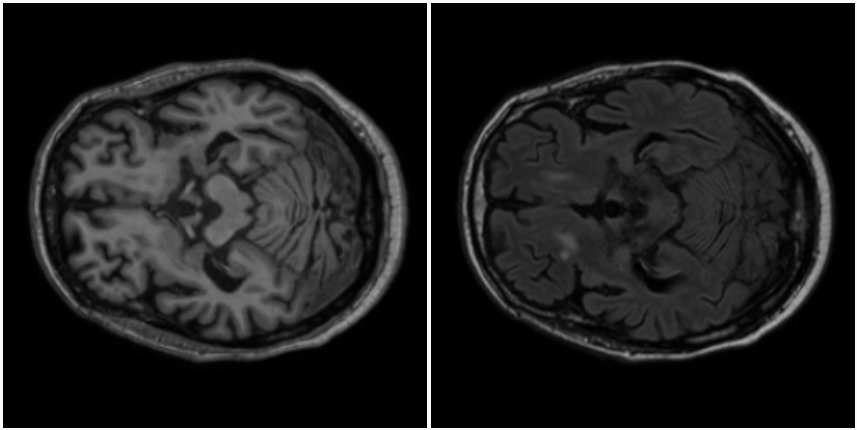
M = number of modalities?

Last time: $x \in \mathbb{R}^{M \times \Omega}$, $x \mapsto^{N_\theta} s_\theta \in \triangle^{K \times \Omega}$

$M = 1$: grayscale, RGB image: $M = 3$



Registration: “align” images



$\mathcal{F}$: fixed image

$\mathcal{M}$: moving image

How to align/overlap them?

Need to find the best transform

$$\mathcal{F} \in \mathbb{R}_1^{\Omega}$$

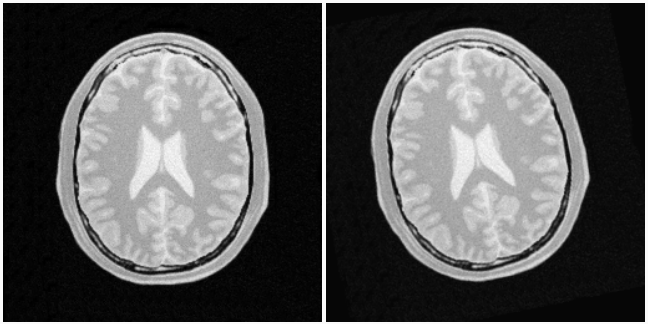
$$\mathcal{M} \in \mathbb{R}_2^{\Omega}$$

$$\text{Transform } \mathcal{T} : \Omega_2 \to \Omega_1$$

$$\min_{\mathcal{T}} \mathcal{L}(\mathcal{F}, \mathcal{T}(\mathcal{M}))$$

$\mathcal{L}$: a loss?

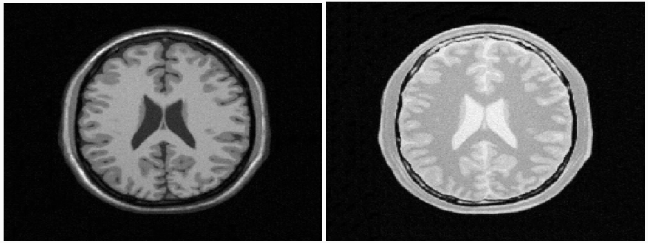
$\mathcal{L} : \Omega_1 \times \Omega_1 \to \mathbb{R}$, "How good" the images are aligned?



$\mathcal{F}$: fixed image $\mathcal{M}$: moving image

L2: $\sum_{i \in \Omega_1} |F^{(i)} - \mathcal{T}(\mathcal{M})^{(i)}|^2$

Different intensities



$\mathcal{F}$: fixed image $\mathcal{M}$: moving image

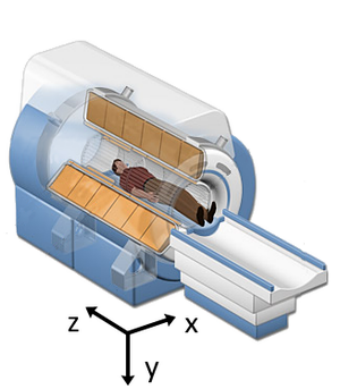
Intensity difference is not meaningful

$$\begin{aligned}\mathsf{MI}(\mathcal{F}, \mathcal{T}(\mathcal{M})) &= \sum_{i,j \in \Omega_1} p(\mathcal{F}^{(i)}, \mathcal{T}(\mathcal{M})^{(j)}) \log \left(\frac{p(\mathcal{F}^{(i)}, \mathcal{T}(\mathcal{M})^{(j)})}{p_{\mathcal{F}}(\mathcal{F}^{(i)}) p_{\mathcal{T}(\mathcal{M}))}(\mathcal{T}(\mathcal{M}))^{(j)})} \right) \\ &= H(\mathcal{F}) + H(\mathcal{T}(\mathcal{M})) - \mathsf{CE}(\mathcal{F}, \mathcal{T}(\mathcal{M}))\end{aligned}$$

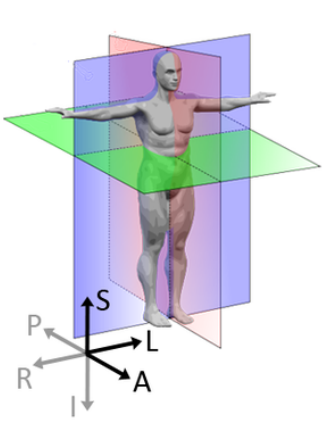
Other losses

	Assumption	Robust to intensity changes	Best for
SSD (Sum of Squared Differences)	Identity relationship between intensities	No	Same-modality images (e.g., CT-CT)
NCC (Normalized Cross-Correlation)	Linear relationship between intensities	Yes (to global changes)	Same-modality images with brightness variations
NGF (Normalized Gradient Fields)	Gradient directions match between images	Yes	Multi-modal images with edge alignment
MI (Mutual Information)	Images share statistical dependencies	Yes	Multi-modal images (e.g., CT-MRI, PET- MRI)
MIND (Modality Independent Neighborhood Descriptor)	Self-similarity patterns are preserved	Yes	Highly different modalities

Coordinates in a scan



World coordinates



Patient coordinates

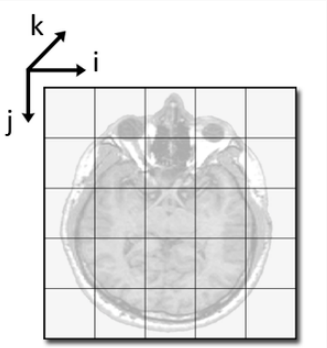


Image coordinates

https://www.slicer.org/wiki/Coordinate_systems

Affine matrix of a scan

Enables to go from image coordinates to patient or world coordinates

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & t_x \\ a_{21} & a_{22} & a_{23} & t_y \\ a_{31} & a_{32} & a_{33} & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Translation

$$\mathrm{T} := \begin{bmatrix} 1 & 0 & 0 & t_x \\ 0 & 1 & 0 & t_y \\ 0 & 0 & 1 & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Scaling

$$\mathtt{S} := \begin{bmatrix} a & 0 & 0 & 0 \\ 0 & b & 0 & 0 \\ 0 & 0 & c & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Also a way to “encode” pixel resolution

Rotation

$$\mathtt{R} := \begin{bmatrix} \cos(\alpha) & -\sin(\alpha) & 0 & 0 \\ \sin(\alpha) & \cos(\alpha) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Shear

$$\mathtt{C} := \begin{bmatrix} 1 & c_x & 0 & 0 \\ c_y & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Putting it all together

$$A = C.R.S.T$$

https://docs.scipy.org/doc/scipy/reference/generated/scipy.ndimage.affine_transform.html

Optimization

Optimize wrt $\mathcal{T}$: find optimal values of $\mathtt{A}$

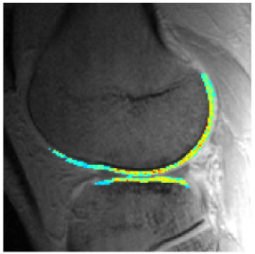
$$\min_{A} \mathcal{L}(\mathcal{F}, \mathcal{T}_A(\mathcal{M}))$$

Example applications (i)

Delayed Gadolinium Enhanced MRI of Cartilage (dGEMRIC)



Damaged cartilage, lower T1

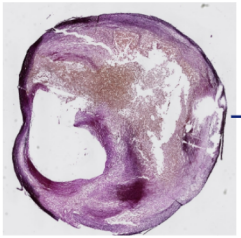


Bron et al, European Radiology, 2013

Example applications (ii)

Multimodal, multiscale registration: histology with carotid CTA

Erasmus MC

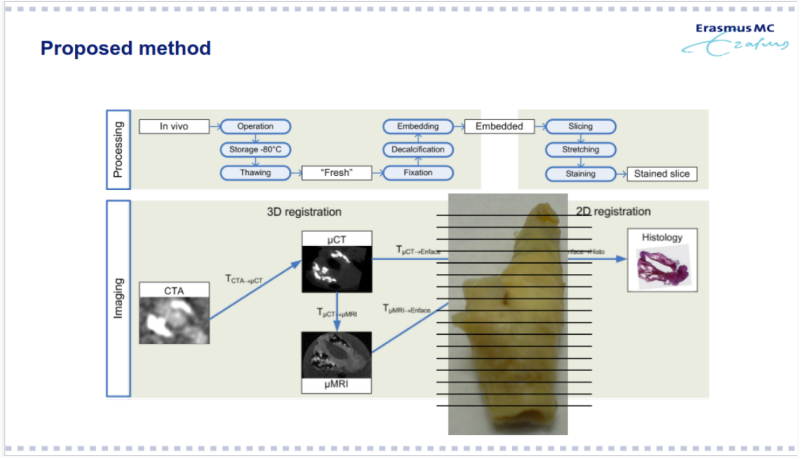


??



Groen et al, Journal of Biomechanics, 2010

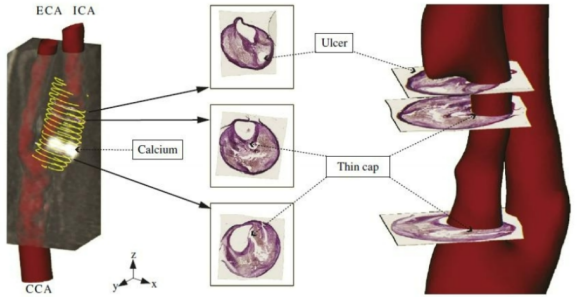
Example applications (ii)



Example applications (ii)

Erasmus MC

Results



Groen et al, Journal of Biomechanics, 2010

2026/09/08 – Registration

Elastic registration

Rigid vs non-rigid

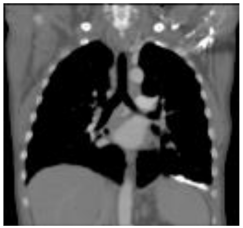
So far: everything rigid, nice for bones and less deformable organs

What about the others?

Need to predict a deformation field

$i \in \Omega_2 \rightarrow \mathbb{R}^3$

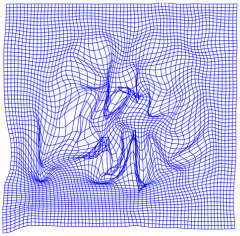
An ill-posed problem



Fixed



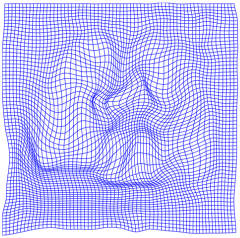
Deformed Moving 1



Deformation 1







Regularizers and folds

Help to constraint the deformatin field

Encourages neighboring pixels/voxels to move in a similar direction (Diffusion regularizer, Curvature regularizer)

$$\min_{\mathcal{T}} \mathcal{L}(\mathcal{F}, \mathcal{T}(\mathcal{M})) + \lambda \mathcal{R}(\mathcal{T})$$

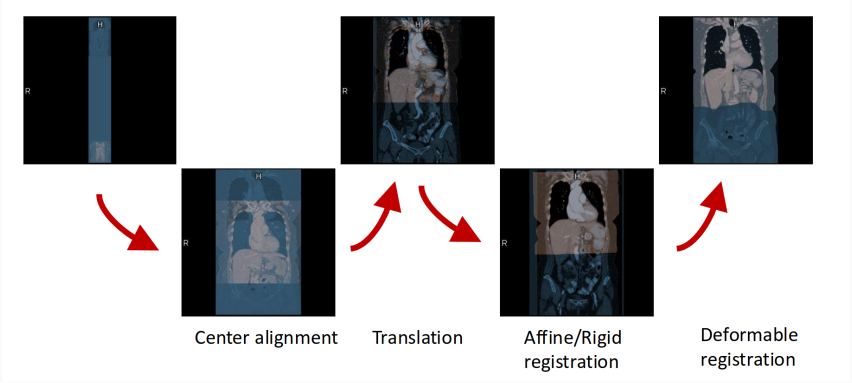
Many optimizers, options, strategies

S. Klein, M. Staring, K. Murphy, M.A. Viergever, J.P.W. Pluim, "elastix: a toolbox for intensity based medical image registration," IEEE Transactions on Medical Imaging, vol. 29, no. 1, pp. 196 - 205, January 2010.

`https://elastix.dev`

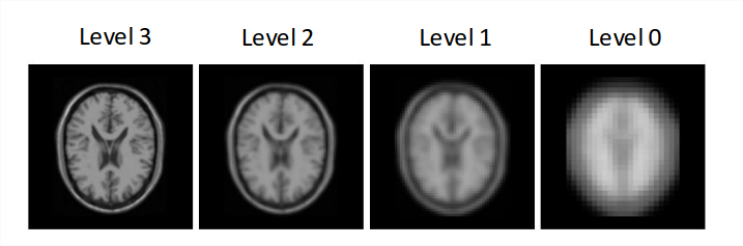
Decompose into sub-steps

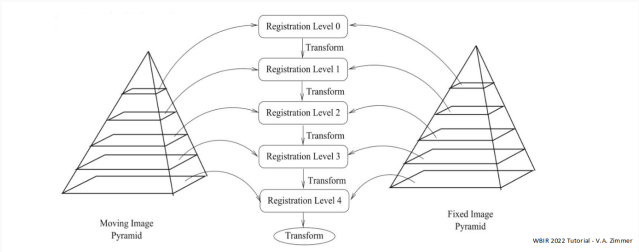
Very ill posed problems, but can limit the difficulties



Author: Alessa Hering

Pyramidal approach





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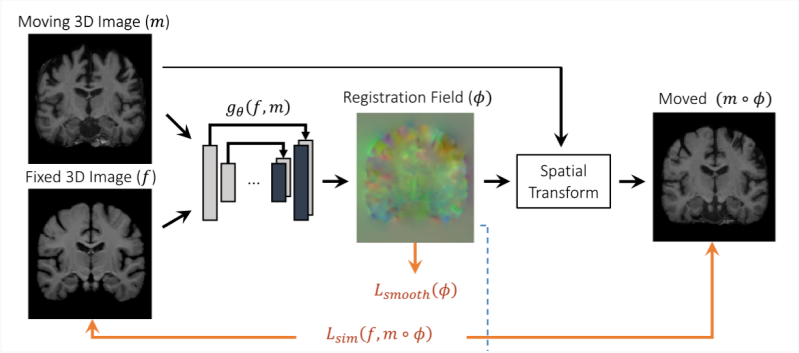
Learned registration

And with deep learning?

Classical: Need to run optimizer for *each* pair of images

Can we train a network instead?

U-Net: never far away



Balakrishnan, G., Zhao, A., Sabuncu, M. R., Guttag, J., & Dalca, A. V. (2019). *Voxelmorph*: a learning framework for deformable medical image registration. IEEE transactions on medical imaging, 38(8), 1788-1800.

$x \mapsto^{R_\theta} t_\theta \in \mathbb{R}^{3 \times \Omega}$

Training data?

How to get the data?

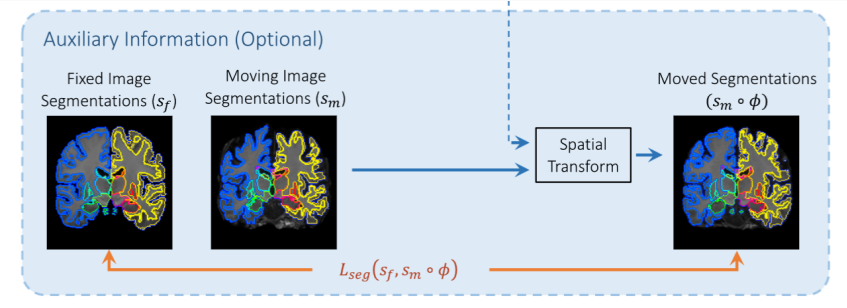
No need! Same objective function, but over a dataset of pairs:

$\mathcal{D} := (\mathcal{F}_n, \mathcal{M}_n)_{n=1}^{N}$

$\min_\theta \sum_{n \in \mathcal{D}} \mathcal{L}(\mathcal{F}_n, t_\theta(\mathcal{M}_n))$

But can use additional regularizers (i)

Segmentation or other landmarks

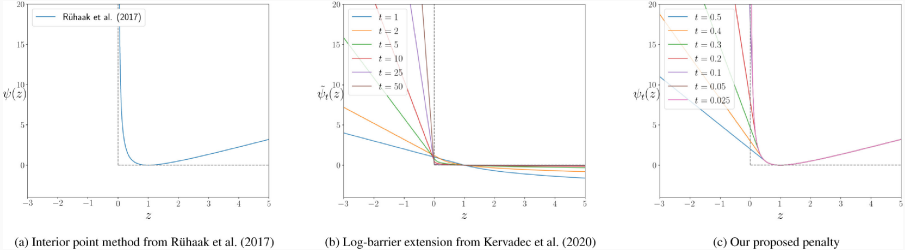


The segmentation must match after transformation

But can use additional regularizers (ii)

Non-fold constraint into the loss

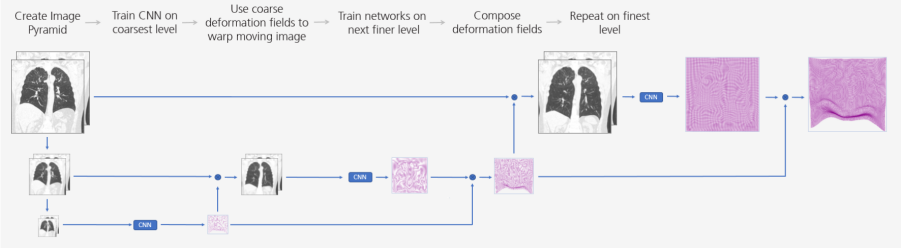
Curvature regularizer $\mathcal{R}$ *must* be greater than 0, and ideally close to 1.



(a) Interior point method from Rühaak et al. (2017) (b) Log-barrier extension from Kervadec et al. (2020) (c) Our proposed penalty

https://doi.org/10.1016/j.media.2021.102139

Multi-resolution strategy also valid



Hering, A., Van Ginneken, B., & Heldmann, S. (2019, October). mlvirnet: Multilevel variational image registration network. InInternational Conference on Medical Image Computing and Computer-Assisted Intervention (pp. 257-265). Cham: Springer International Publishing

Challenge: Learn2Reg



https://learn2reg.grand-challenge.org/

Registration in the Netherlands

Elastix:

Stefan Klein, Marius Starring: now at Erasmus MC (Rotterdam) and LUMC (Leiden) respectively

Josien Pluim: TU/e

Learn2Reg:

Alessa Hering: Radboud UMC

Voxel-morph: Mert Sabuncu, Adrian Dalca: Cornell, MIT & Harvard Medical School

Mutual information for registration: William M. Wells III (Sandy), Harvard Medical School